

CENTRE & SPREAD

Mean $\bar{x} = \Sigma x / n$ =AVERAGE(range)

Median = middle value =MEDIAN(range)

Even $n \rightarrow$ average the two middle values.

$S^2 = \Sigma(x-\bar{x})^2 / (n-1)$ =VAR(range)

$S = \sqrt{S^2}$ =STDEV(range)

Excel	Divides by	Use
STDEV / STDEV.S	$n - 1$	Course default (a sample)
STDEVP / STDEV.P	n	Only if data IS the population

Outliers: the **mean** is very sensitive, the **median** is resistant. Bill Gates walks into the bar $\rightarrow$ mean explodes, median barely moves.

Stdev has the same units as the data. Variance does not.

Z-SCORE — the master tool

$z = (x - \text{mean}) / \text{stdev}$

Distance from the mean measured in **stdev units**.

Unusualness = how many stdevs from the mean. Compare $|z|$, never raw values.

Works for ANY distribution, not just Normal.

Shanghai	NYSE
$x=11.75$, $\mu=0.57$, $\sigma=7.25$	$x=10.39$, $\mu=0.46$, $\sigma=4.49$
$z = 1.54$	$z = 2.21 \leftarrow$ more exceptional

Higher raw return, yet the smaller move was the rarer event.

Z $\rightarrow$ PERCENTILE (needs Normality)

=NORMSDIST(z) $\rightarrow$ % BELOW x

=NORMDIST(x, μ , σ , TRUE) (same thing)

=NORMSINV(p) $\rightarrow$ z for a given p

Example. Score 90, $\mu=75$, $\sigma=10$

$z = (90-75)/10 = 1.5 \rightarrow$ NORMSDIST(1.5) = **0.933** $\rightarrow$ 93rd percentile.

If instead $\mu=80$: $z = 1.0 \rightarrow$ **0.841** $\rightarrow$ only 84th.

Raw scores mean nothing alone. **The mean and stdev are everything.**

EMPIRICAL RULE

Interval	Inside	Outside
$\mu \pm 1\sigma$	68.3%	31.7%
$\mu \pm 2\sigma$	95.4%	4.6%
$\mu \pm 3\sigma$	99.7%	0.3%

Weight $\mu=180$, $\sigma=10 \rightarrow$ 95.4% of men weigh 160–200 lbs.

SKEWNESS — asymmetry

=SKEW(range) $\approx 3(\text{Mean}-\text{Median})/\text{Stdev}$

Skew	Order	Meaning
+ right	Mode < Med < Mean	A few very LARGE values
0	Mean = Med = Mode	Symmetric (Normal)
– left	Mean < Med < Mode	A few very SMALL values

|skew| ≥ 0.5 is significant.

Mean > Median $\Rightarrow$ right skew. Income, market caps, movie revenues.

Mean < Median $\Rightarrow$ left skew. Easy test grades, satisfaction scores. *Skew -3.2 on satisfaction = most are happy, a small tail is very unhappy.*

KURTOSIS — tail weight

TRAP: Excel =KURT() returns **EXCESS** kurtosis (Normal = 0). The lecture scale has Normal = **3**. **Add 3 to Excel's answer.**

High kurtosis = **heavier tails** = extreme events (*both* directions) happen more often than Normal predicts. *Call centre: agent A kurt 3 = Normal; agent B kurt 5 $\rightarrow$ more very-long AND very-short calls, so the Normal approximation is riskier for B.* S&P 500 monthly: $\mu=0.66\%$, $\sigma=3.5\%$. March 2020 was a **-6σ** event — essentially impossible under Normality, yet it happened.

CORRELATION

=CORREL(Xrange, Yrange) $-1 \leq \text{Corr} \leq 1$

$\text{Cov} = \text{Corr} \times \sigma_x \times \sigma_y$ | $\text{Corr} = \text{Cov}/(\sigma_x \cdot \sigma_y)$

Value	Meaning
+1	Perfect positive line
+	Same direction
0	No linear relationship
–	Opposite directions
–1	Perfect negative line

Sign = direction. **Magnitude** = strength (tightness around a line).

THE FOUR CORRELATION TRAPS

1. Corr = 0 $\neq$ independent. A circle, a parabola, a wave all give Corr = 0 while being perfectly dependent. It only detects **straight lines**.

2. Corr $\neq$ slope. Corr 0.30 does NOT mean "30 cents per dollar". Tight clustering $\neq$ steep line.

3. Corr $\neq$ causation. Foot size vs education = 0.849.

4. Correlations break down in crises. LTCM 1998: Russia defaulted, historical relationships inverted, $-\$4.6\text{bn}$ in four months.

T/F drill (Corr = 0.30, restaurants vs fitness): positive linear relationship ✓ · below-average on one $\Rightarrow$ expect below-average on other ✓ · "30 cents per dollar" ✗ · proves causation ✗

POPULATION vs SAMPLE

	Population	Sample
Size	N	n
Mean	μ	$\bar{x}$
Stdev	σ	S
Proportion	p	$\hat{p}$
Essence	Constant	Random variable
Known?	Never	Yes, after sampling

LECTURE QUIZ ANSWERS

Q. Mean market cap $\gg$ median. What does that say about skew?

A. Right / positive skew — a few giant firms drag the mean above the bulk of companies.

Q. Satisfaction scores have skew -3.2 . Which is true?

A. Most customers are satisfied, but a small number are very dissatisfied (long LEFT tail).

Q. Two agents, same mean (6 min) and stdev (2 min), skew ≈ 0 , but kurtosis 3 vs 5.

A. Agent B has **more extremely long AND extremely short** calls. A Normal distribution has kurtosis 3, so **the Normal approximation is riskier for B** — it will understate the chance of a 15-minute call.

HISTOGRAM & REAL-WORLD SHAPES

Data	Shape
US household income	Right skew (mean \$106.5k > median \$83.7k)
Movie box office	Strong right skew (Minecraft $z = 3.4$)
Easy exam grades	Left skew
Stock returns	$\approx$ symmetric but heavy tails

Right-skewed data $\rightarrow$ quote the **median** as "typical", not the mean.

EXPECTED VALUE & VARIANCE OF A RV

$$E(X) = \mu = \sum x \cdot P(X=x)$$

$$V(X) = \sigma^2 = \sum P(X=x) \cdot [x - E(X)]^2$$

$$\sigma = \sqrt{V(X)}$$

E(X) is: probability-weighted average · long-run average · fair value · balance point.

V(X) is **never negative**. If $V(X)=0$ then X is a constant, not a RV.

Worked — contractor sells 0/1/2/3 flats, $P = .1/.3/.3/.3$

$$E(X) = 0(.1) + 1(.3) + 2(.3) + 3(.3) = 1.8$$

$$V(X) = .1(0-1.8)^2 + .3(1-1.8)^2 + .3(2-1.8)^2 + .3(3-1.8)^2 = 0.96 \rightarrow \sigma = 0.9797$$

LINEAR TRANSFORM $Y = a + bX$

$$E(a+bX) = a + b \cdot E(X)$$

$$V(a+bX) = b^2 \cdot V(X)$$

$$\text{stdev}(a+bX) = |b| \cdot \text{stdev}(X)$$

Note: the constant **a** shifts the mean but **NEVER** changes the spread.

Same contractor: profit $Y = -50 + 200X$ (\$K)

$$E(Y) = -50 + 200(1.8) = 310 \cdot V(Y) = 200^2(0.96) = 38,400 \cdot \sigma(Y) = 200(0.9797) = 195.96$$

Non-linear payoff (e.g. a bonus 0/15/52.5/82.5)? You CANNOT use this rule — rebuild the table and take $\sum g(x) \cdot P$. Gives $E=45$, $\sigma=30.187$.

PROBABILITY TREES

ALONG a path (AND) → MULTIPLY
BETWEEN paths (OR) → ADD

"At least one" over N periods

$$P(\text{never}) = (1 - p)^N$$

$$P(\text{at least one}) = 1 - (1 - p)^N$$

AAA bond, $p = 0.0002/\text{yr}$, 25 yrs:

$$1 - (0.9998)^{25} = 0.004988 \approx 0.5\% \text{ Two dice: 36 equally likely paths. } E(\text{sum})=7. \text{ Chuck-a-luck: } E = -\$0.0463, \sigma = 1.288 \text{ — all casino games are bad bets.}$$

BAYES / REVERSE PROBABILITY

$$P(A|+) = P(A) \cdot P(+|A) \div P(+)$$

$$P(+) = P(A)P(+|A) + P(\text{not}A)P(+|\text{not}A)$$

HIV worked example

$$P(\text{HIV}) = .006 \cdot \text{sensitivity} = .999 \cdot \text{false alarm} = .01$$

$$P(+") = .006(.999) + .994(.01) = .015934$$

$$P(\text{HIV} | "+") = .005994 / .015934 = 0.38$$

A 99.9% accurate test for a **rare** condition still gives mostly **false alarms** — because the healthy group is so much larger.

★ SAXBY — THE KEY FORMULA ★

$$E(aX + bY) = a \cdot E(X) + b \cdot E(Y)$$

$$V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2ab \cdot \sigma_X \cdot \sigma_Y \cdot \text{Corr}(X, Y)$$

$$\text{stdev} = \sqrt{V}$$

Question	a	b
Total $X+Y$	1	1
Difference / $P(X>Y)$	1	-1
Portfolio w in X	w	$1-w$
Weighted 35/65	0.35	0.65

TRAP: stdev is **NEVER** just added.

$\text{stdev}(aX+bY) \leq a \cdot \sigma_X + b \cdot \sigma_Y$, with equality **only when** $\text{Corr} = +1$. That gap is diversification.

Cross term signs: $\text{Corr} > 0$ raises risk · $\text{Corr} = 0$ kills the term · $\text{Corr} < 0$ lowers it. With **b negative** the term **subtracts**.

PORTFOLIO $P = wX + (1-w)Y$

$$E(P) = w \cdot \mu_X + (1-w) \cdot \mu_Y$$

$$V(P) = w^2 \sigma_X^2 + (1-w)^2 \sigma_Y^2 + 2w(1-w) \sigma_X \sigma_Y \cdot \text{Corr}$$

① **TARGET RETURN → solve for w (linear)**

$$w = (\text{target} - \mu_Y) / (\mu_X - \mu_Y)$$

$$\mu_X=10\%, \mu_Y=20\%, \text{target } 15\% \rightarrow w = 0.5 \text{ (50/50)}.$$

② **TARGET RISK → solve for w (quadratic)**

$$A = \sigma_X^2 + \sigma_Y^2 - 2 \cdot \text{Corr} \cdot \sigma_X \cdot \sigma_Y$$

$$B = 2 \cdot \text{Corr} \cdot \sigma_X \cdot \sigma_Y - 2 \sigma_Y^2$$

$$C = \sigma_Y^2 - \text{target}^2$$

$$w = (-B \pm \sqrt{B^2 - 4AC}) / 2A$$

$$\sigma_X=15\%, \sigma_Y=35\%, \text{Corr}=0.5, \text{target } 20\% \rightarrow w \approx 0.60 \text{ (take the root inside } [0,1]).$$

③ **MINIMUM-RISK PORTFOLIO**

$$w^* = (\sigma_Y^2 - \text{Corr} \cdot \sigma_X \cdot \sigma_Y) / (\sigma_X^2 + \sigma_Y^2 - 2 \cdot \text{Corr} \cdot \sigma_X \cdot \sigma_Y)$$

$$\text{Real estate } \sigma=3.16 \text{ \& } 2.68, \text{Corr}=-0.5 \rightarrow w^*=0.44, \text{min } \sigma = 1.45 \text{ — below BOTH assets.}$$

DIVERSIFICATION — WHAT TO SAY

Corr	Effect on the frontier
+1	No benefit at all. Straight line.
+	Mitigates the portfolio effect
0	Cross term vanishes; real reduction
-1	Maximum; risk can reach zero

The lower the Corr, the more the curve **bends left**.

The portfolio effect: a diversified portfolio can have **lower volatility than any single asset**. Why? Idiosyncratic randomness averages out.

Efficient frontier = upper-left boundary = highest return for each level of risk (Markowitz, 1952).

FEASIBILITY DRILL ($0 \leq w \leq 1$)

Can the portfolio...

...return more than the better asset?	NO
...return less than the worse asset?	NO
...have stdev above the riskier one?	NO*
...have stdev BELOW the safer one?	YES

*Return is always a weighted average, so it is bounded by the two assets. Risk is NOT — that asymmetry is the whole point. *Unless Corr is high and weights extreme.*

RESOURCE POOLING

Capacity to cover k stdevs: $\mu + k \cdot \sigma$

$$\text{Separate: } (\mu_X + k\sigma_X) + (\mu_Y + k\sigma_Y)$$

$$\text{Pooled: } (\mu_X + \mu_Y) + k \cdot \sigma(X+Y)$$

GPUs: $\mu=20$, $\sigma=10$ each, $\text{Corr}=0.2$, $k=3$

Separate: $20+30 = 50$ each $\rightarrow 100$

$$\text{Pooled: } \sigma = \sqrt{(100+100+2 \cdot 10 \cdot 10 \cdot 0.2)} = 15.49 \rightarrow 40+3(15.49) = 86.5$$

Saves ≈ 13.5 GPUs. Peaks in one cancel troughs in the other.

THE FOUR NORMAL QUESTIONS

$X \sim N(\mu, \sigma^2)$. Continuous, so $P(X = x) = 0$. Probability = area under the curve. **NORMDIST** always gives the area to the **LEFT**.

#	Question	Excel
1	$P(X < x)$	=NORMDIST(x, μ , σ , TRUE)
2	$P(X > x)$	=1-NORMDIST(x, μ , σ , TRUE)
3	$P(a < X < b)$	=NORMDIST(b, μ , σ , TRUE) -NORMDIST(a, μ , σ , TRUE)
4	Reverse: find x	=NORMINV(p, μ , σ)

Worked — grades $\mu=80$, $\sigma=10$

$P(X < 90) = \text{NORMSDIST}(1) = \mathbf{0.8413}$

$P(X > 90) = 1 - 0.8413 = \mathbf{0.1587}$

$P(75 < X < 90) = 0.8413 - 0.3085 = \mathbf{0.5328}$

REVERSE — GETTING p RIGHT

TRAP: "Top 10%" means $P(X < x) = \mathbf{0.90}$.

Use **NORMINV(0.90, ...)**, not 0.10.

Asked for	Feed NORMINV
Top 10% (High Pass)	0.90 → 92.8
Top 5%	0.95
Bottom 5% (worst loss)	0.05
Service level 99%	0.99

VaR: portfolio $\mu=7\%$, $\sigma=10\%$, worst 5%:

NORMINV(0.05, 7, 10) = -9.45% Common z: 90% → 1.28 · 95% → 1.645 ·

97.5% → 1.96 · 99% → 2.33

★ SAXBY + NORMAL = THE BIG ONE ★

STEP 1 — build the combination with SAXBY: get the new **mean** and new **stdev**.

STEP 2 — treat it as ONE normal variable and answer with **NORMDIST** / **NORMINV**.

Worked — retail demand (Lec 5, Ex 3)

$X \sim N(100, 10^2)$ walk-in, $Y \sim N(150, 15^2)$ phone, $\text{Corr} = 0.8$

Part	Work
Total $X+Y$	$E=250$ $\sigma=\sqrt{(100+225+2 \cdot 10 \cdot 15 \cdot 0.8)}=\mathbf{23.77}$
Stock for 99%	NORMINV(.99, 250, 23.77)=305.3 → 306
$P(X > Y)$	$X-Y$: $E=-50$, $\sigma=\sqrt{(100+225-240)}=\mathbf{9.22}$ $P(X-Y > 0)=1-\text{NORMDIST}(0, -50, 9.22, 1)$ $\approx \mathbf{0}$

$P(X > Y) \Rightarrow P(X - Y > 0)$. Set $a=1$, $b=-1$, then ask about **zero**. Note the cross term now **subtracts**.

Inventory answers must be ROUNDED UP to whole units.

WEIGHTED SCORES (Gifted & Talented)

SAI 35%, NAI 65%, both $N(100, 16^2)$. Ann: 119 and 123 → weighted **121.6**.

Corr	$\sigma(\text{Agg})$	Percentile
1.0	16.0	0.9115
0.5	14.1	0.937

Lower correlation $\Rightarrow$ smaller $\sigma \Rightarrow$ the same score lands at a HIGHER percentile.

EXAM PLAYBOOK — decision tree

If the question gives you...	Do
A list of raw numbers	AVERAGE / STDEV / SKEW
x , μ , $\sigma \rightarrow$ "what %?"	z then NORMSDIST
A % $\rightarrow$ "what score?"	NORMINV
Outcomes + probabilities	$\Sigma x \cdot P$ and $\Sigma (x-E)^2$
$Y = a + bX$	$a + bE(X)$, $ b \cdot \sigma$
Multi-step / "at least one"	Tree: $1 - (1-p)^N$
"Given the test was +"	Bayes
Two RVs and a Corr	SAXBY
Two RVs + a probability	SAXBY then NORMDIST
Weights / target return / risk	Portfolio formulas

FINAL CHECKLIST

- Did I use STDEV (sample), not STDEVP?
- KURT needs **+3** for the lecture scale.
- "Greater than" $\rightarrow 1 - \text{NORMDIST}$.
- "Top p%" $\rightarrow$ feed **NORMINV 1 - p**.
- For $X - Y$, **b = -1**; cross term subtracts.
- Did I take the $\sqrt{\quad}$ at the end? Variance $\neq$ stdev.
- μ and σ on the same scale (both % or both decimal).
- Inventory / units $\rightarrow$ **round UP**.
- Two roots on target-risk? Pick the one in [0,1].
- Show the formula in Canvas for partial credit.

STANDARD NORMAL $P(Z < z)$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Negative z: $P(Z < -z) = 1 - P(Z < z)$.